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PAPER_1008: Production Scaling v14 — 600k calc/s

Abstract

We upgrade the production benchmark from v13 (550k calc/s, 20 kernels) to v14 (600k calc/s, 24 kernels). Four new kernels using $S_{26}^{(3)}$:

Kernel	Description
kernel_agn_merger_fubi	AGN merger F_U_Bi with $S_{26}^{(3)}$
kernel_qgp_vacuum_density	QGP vacuum density $\rho_{\text{QGP}}(T)$
kernel_alice_multiplicity	ALICE $dN_{\text{ch}}/d\eta$ SCm scaling
kernel_ym_mass_gap	Yang-Mills Δ_{YM} via BCS phonon

1. Scaling History

Version	Target (calc/s)	Kernels
v11	500,000	16
v13	550,000	20
v14	600,000	24

2. REST API

20 total routes including new endpoints: - POST /api/fubi/agn-merger — AGN merger F_U_Bi with $S_{26}^{(3)}$ - POST /api/qgp/scm-dynamics — QGP SCm phonon dynamics

3. Implementation

File: production_scaling_v14.py, class ProductionScalingV14. CP4 class #592.